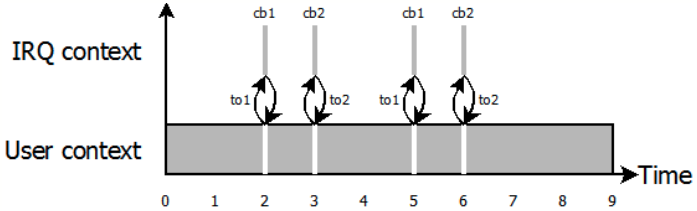
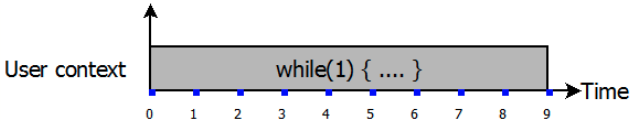
# Mbed Interrupts
# Timeout Interrupt – Formative Assessment (FA1)

- A rule of thumb: Don't use *wait()* in ISR call backs (*cb*).
- The implementation we made was for the illustration of the concept only.
- **You are now required to write a code that implements two timeouts to achieve the following behaviour:**
  - led1: flash brief one per second, EVERY second.
  - led2: On at t = 2 and off at t = 5
  - led3: On at t = 3 and off at t = 6

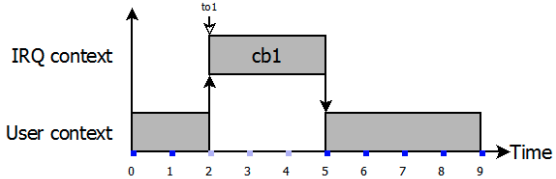



- Solution will be uploaded to Moodle after **two weeks**.

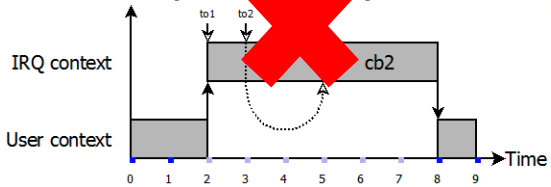
## Without the Timeout (to)



## With one Timeout (to1)




## With two Timeouts (to1 & to2)



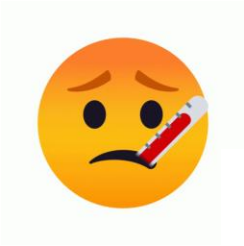

## Timeout (to)

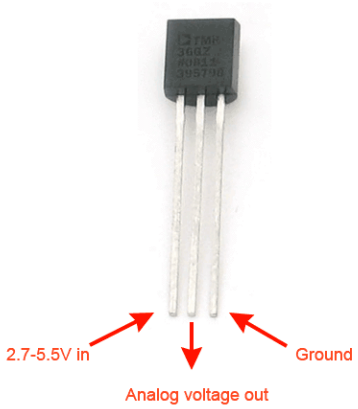
# ADC Implementation – Digital Thermometer (FA2)

- We are required to write a C/C++ code to create a digital thermometer using the TMP36 that displays the Temp value in ºC on the serial terminal.
- The output voltage of TMP36 is linearly proportional to the measured temperature.

$$T = (V_0 - 500)/10$$

```
//Digital Thermometer
```





# ADC Implementation – Digital Thermostat (FA3)

- We are required to write a C/C++ code to create a digital thermostat using the TMP36 that displays the Temp value in $^{o}C$ on the serial terminal.
- The **user enters the desired Max and Min temperatures** from the keyboard. The program measures the ambient temperature every second. If the temperature is above the Max value then the message ***"ALARM-HIGH TEMPERATURE"*** is displayed and the ***User LED is lit***. Also, if the temperature is below the Min value then the message ***"ALARM-LOW TEMPERATURE"*** is displayed and the ***LED is lit*** to indicate the alarm condition. If the temperature is within the desired limits then the message "***NORMAL TEMPERATURE"*** is displayed and the ***LED is turned OFF***.